Calculation of the PACF : m = 2

Let $\underline{\beta} = \begin{pmatrix} \beta_{2,1} \\ \beta_{2,2} \end{pmatrix}$; $P_2 = \begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix}$; $\underline{\rho}_2 = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$

PACF at lag 2 $(m=2)$

We solve the following for $\underline{\beta}$

$$P_2 \underline{\beta} = \underline{\rho}_2$$

$$\Rightarrow \underline{\beta} = P_2^{-1} \underline{\rho}_2$$

$$= \begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$$

$$= \frac{1}{1-\rho_1^2} \begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$$

$$= \frac{1}{1-\rho_1^2} \begin{pmatrix} \rho_1 - \rho_1 \rho_2 \\ \rho_2 - \rho_1^2 \end{pmatrix}$$

$$\beta_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}.$$

$$\rho(0,2|1) = \frac{\rho_2 - \rho_1^2}{1-\rho_1^2}.$$